

RESEARCH ARTICLE

When does extreme drought elicit extreme ecological responses?

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Abstract

1. Global climate change models predict an increase in the frequency and intensity of extreme droughts, with uncertain ecological impacts across ecosystems. In particular, it is not clear when extreme droughts will elicit extreme ecological responses.
2. For this study, we employed three complementary approaches to explore the relationships between extreme drought and ecosystem responses. First, we used global data mining to evaluate the relationship between extreme gross primary productivity and extreme precipitation from 1980 to 2013. Second, we conducted a meta-analysis using 132 drought experiments across the globe to assess the response ratios (RRs) of above-ground net primary productivity (ANPP) to extreme versus non-extreme drought treatments. Third, we examined community and ecosystem responses in an alpine meadow to a drought gradient experiment, which included five precipitation treatment levels (1/12 P, 1/4 P, 1/2 P, 3/4 P, and P, where P is the growing season precipitation).
3. This study had three key results. In our historical data mining, we found that extreme droughts elicited extreme ecological responses only 15.1% of the time. The meta-analysis results indicated that there were no significant differences in the RRs of ANPP between the extreme versus non-extreme drought treatments. The drought gradient experiment results revealed that although the four drought treatments were statistically extreme, only the most extreme drought treatment (1/12 P) significantly reduced ANPP over the 3 years. Meanwhile, species richness and asynchrony were significantly reduced under the 1/12 P treatment, which led to a significant reduction in productivity.
4. *Synthesis.* These results suggest that extreme ecological responses to extreme drought may be less frequent than previously thought. But when they do occur, extreme ecological responses may be driven by plant community changes such as species asynchrony, species loss or species reordering. Our experimental results highlight the key role of community dynamics in determining the resistance of ecosystem productivity to extreme drought, which should be assessed when predicting ecological responses to climate change.

KEYWORDS

climate extreme, data mining, extreme response, gradient experiment, meta-analysis, resistance, species asynchrony, species richness

1 | INTRODUCTION

Extreme droughts can trigger abnormally severe impacts on both natural and human systems (Ciais et al., 2005; Reichstein et al., 2013; Schwalm et al., 2012) and are projected to continue to increase in frequency and intensity over the next century (Diffenbaugh, Swain, & Touma, 2015; Meehl et al., 2000; Sergio et al., 2014; Sheffield & Wood, 2008; Wuebbles et al., 2014). Many experimental and observational studies have reported that extreme droughts reduced primary productivity (Ciais et al., 2005; Knapp et al., 2008; Liu, Ogaya, Barbeta, Yang, & Peñuelas, 2015). In contrast, other studies found that productivity remained remarkably stable under such extremes (Jentsch et al., 2011; Kreyling, Beierkuhnlein, et al., 2008; Kreyling, Wenigmann, Beierkuhnlein, & Jentsch, 2008). There are several possible explanations to such divergent responses to drought. First, differences in the duration and/or magnitude of drought across the various experiments and observations may cause variation in ecological responses. Second, community-level responses (e.g. complementary responses between species) can buffer primary production without altering community composition (Jentsch et al., 2011). Third, ecological legacies from previous extreme events may influence responses to extreme droughts, and further complicate comparisons across ecosystems (Anderegg et al., 2012; Kaisermann, Vries, Griffiths, & Bardgett, 2017; Laura Suarez & Kitzberger, 2010; Petrie et al., 2018; Wu et al., 2018). Overall, we still lack a comprehensive understanding of the mechanisms driving the responses of ecosystems to extreme drought.

Extreme drought is typically quantified as precipitation below the 5th or 10th percentile of the historical probability distribution function (Figure S1) (Hoover, Knapp, & Smith, 2014b; von Buttlar et al., 2018; Zscheischler, Mahecha, Harmeling, & Reichstein, 2013). Similarly, an ecological response (e.g. gross primary productivity [GPP]) can be classified as extreme if it below a certain threshold. However, extreme ecological responses have rarely been addressed and remain highly uncertain relative to climate extremes (Liu, Liu, & Yin, 2013). Smith (2011) defined 'extreme climatic events' as climate extremes that elicit an extreme ecological response, which combines both climate drivers and ecological responses in a synthetic definition. Nevertheless, the correlations between extreme drought and extreme ecosystem function remain highly illusive. We hypothesized that extreme drought may not necessarily cause extreme ecosystem responses.

Extreme ecosystem responses to drought are predicted to occur when a key threshold is crossed and individual-level effects are amplified to higher hierarchical domains, resulting in significant changes in community structure and ecosystem function (Felton & Smith, 2017; Niu et al., 2014; Smith, 2011). Thus, the magnitude of extreme drought effects on ecosystem function may be contingent on whether, and how, plant community composition is altered under the changing availability

of resources (Bloor & Bardgett, 2012; Brockway, 1998; Smith, Knapp, & Collins, 2009). Species diversity is expected to drive changes in ecosystem productivity (García-Palacios, Gross, Gaitán, & Maestre, 2018; Isbell, Polley, & Wilsey, 2009; Pennekamp et al., 2018), as biodiversity can stabilize ecosystem productivity by increasing its resistance to climate change (Isbell et al., 2015). Even if species richness is stable, community composition may be altered through reordering of species, the loss of dominant species stability or species asynchrony (Hoover et al., 2014b; Xu et al., 2014). Thus, changes in community structure are predicted to be a key mechanism driving ecological responses to climate extremes (Smith, 2011), yet empirical support is still lacking. We hypothesized that extreme ecological response to extreme drought would most likely occur if species and their asynchrony are reduced.

The goals of this paper are to (a) detect whether historical extreme drought, or experimental extreme drought cause extremes response in ecosystem productivity, (b) examine what magnitude of drought elicits extreme ecological responses, and (c) investigate the biological mechanisms for extreme ecological responses. To achieve these goals, we conducted three complementary studies. The first study investigated the relationship between observed global historical drought and GPP. These data, gathered by satellites and models, extend back to the 1970s, enabling the study of extreme historical events over many decades. The second study was a global meta-analysis that used data from published drought experiments to explore whether extreme drought treatments might cause extreme ecosystem responses. The third study was a site-based, drought gradient experiment, which imposed a precipitation reduction gradient, with the objective of investigating the mechanisms driving extreme ecosystem responses to extreme drought. The three studies combined here were complementary, allowing for a more comprehensive exploration of the relationship between extreme drought and ecosystem responses.

2 | MATERIALS AND METHODS

2.1 | Historical data mining

We analysed global GPP data from the FLUXCOM global carbon flux dataset. These data were estimated by three machine learning methods, which were trained on daily carbon flux data from 224 flux tower sites to produce spatiotemporal carbon flux grids (Jung et al., 2017). We calculated mean annual GPP and extreme annual GPP (<5th or 10th percentile) (Hoover et al., 2014b), with grids set at a 0.5° (c. 56 km) spatial resolution for the 1980–2013 period, and the distribution of annual GPP for each grid spanning these years based on the 34-year historical record.

Our global gridded precipitation data were obtained from the CRU TS v. 4.01 dataset (Harris, Jones, Osborn, & Lister, 2014) and spanned

the period from January 1980 to December 2013. The CRU time series consisted of total monthly precipitation at a high-resolution grid ($0.5 \times 0.5^\circ$, c. 56×56 km at the equator), which was based on measurements from globally distributed weather stations. As with GPP, we categorized extreme droughts as years with annual precipitation lower than 5th or 10th percentile of the historical probability distribution (Hoover et al., 2014b; von Buttler et al., 2018; Zscheischler et al., 2013). We also identified droughts using the Standardized Precipitation-Evapotranspiration Index (SPEI), with extreme droughts as years with an SPEI < -2 (Ahmadali-pour, Moradkhani, & Svoboda, 2017; Begueria & Vicente Serrano, 2017; Potop, Možný, & Soukup, 2012; Smirnov et al., 2016; Tao, Borth, Fraedrich, Su, & Zhu, 2014).

We focused on the correlations of three conditions, (a) extremely low GPP and extreme drought occurring during the same year, (b) extremely low GPP lagging 1 year behind extreme drought, and (c) extremely low GPP lagging 2 years behind extreme drought. We analysed the responses of extremely low GPP to extreme drought at both the 5th and 10th percentiles, by calculating their correlation times. For example, if extreme drought occurred twice in one grid over the 34 years, each of these extreme drought events potentially led to extremely low GPP in the current year or lagged 1 or 2 years behind. If extremely low GPP occurred during the same year as extreme drought, we scored one correlation in the same year. If extremely low GPP occurred after 1 or 2 years, we recognized this as 1 or 2 year lagging correlation. Finally, we calculated the percentage of correlation in extreme low GPP as below.

$$R_c = \sum H_i / H, \quad (1)$$

where H is the total extremely low GPP events that were potentially caused by extreme drought. Theoretically, we assumed that one extreme drought may cause extremely low GPP during the same year, or lagged 1 or 2 years behind. H_i is the extremely low GPP events that were driven by extreme drought (during the same year or lagged 1 or 2 years behind). For example, if extreme drought caused extremely low GPP during the same year and also lagged by 1 year, then $H_i = 2$.

2.2 | Global meta-analysis

We searched for published experimental studies over the period 1992–2016 on Web of Science by using a variety of keywords (precipitation experiment, drought, net primary productivity). We then filtered the search results according to the following criteria: (a) the control and precipitation treatment plots were conducted in the same sites and time with the same ecosystems, and (b) the experiments were field-based. For multifactor experiments, we treated the plots under other factors as independent measurements (e.g. warming, N addition). The compiled dataset included soil moisture and above-ground net primary productivity (ANPP). We also obtained mean annual precipitation, mean annual temperature and precipitation reduction levels at each site from the publications. Control and treatment mean, standard deviations or standard errors and sample sizes of each variable were directly extracted or indirectly calculated from the articles. We used GetData Graph Digitizer 2.24 (available

online: <http://getdata-graph-digitizer.com>) to extract the numerical values from the published figures. Our final data set consisted 132 drought experiments and included forests, grasslands and shrublands (Figure S2, Table S1). The mean annual precipitation of those studies ranged from 163 to 2,000 mm (Figure S3), and experimental duration ranged from 1 to 15 years (Figure S4).

We used effect size to quantify the response of carbon cycle variables to global rainfall manipulation treatments. The effect size of each variable was calculated by the nature log of response ratio (RR):

$$RR = \ln(\bar{X}_t / \bar{X}_c) \quad (2)$$

with a variance of:

$$\text{var}(RR) = SD_t^2 / (N_t X_t^2) + SD_c^2 / (N_c X_c^2) \quad (3)$$

and where $\bar{X}_t$, SD_t and N_t are the mean, standard deviation and sample size of a response variable in treatment plots; $\bar{X}_c$, SD_c and N_c represent the same variable in control plots, respectively (Hedges, Gurevitch, & Curtis, 1999).

In the meta-analyses of global precipitation experiments, we could not determine the extremity of the treatment (e.g. 5th percentile) because the experiments did not provide historical precipitation. However, according to the data mining study, the 1st, 5th and 10th percentiles of annual precipitation were on average 36.2%, 29.4%, and 24.4% reductions of mean annual precipitation respectively (Figure S5). Using the 5th percentile, we separated the treatments into three groups, 0%–30%, 30%–60%, and >60% precipitation reduction, and presumed that the last two groups were likely extreme drought treatments. We also separated the treatments into three (0%–40%, 40%–80%, and >80%) and five (0%–20%, 20%–40%, 40%–60%, 60%–80%, and >80%) precipitation reduction groups based on the 1st and 10th percentiles thresholds respectively. This allowed us to compare the RRs between extreme versus non-extreme treatments.

2.3 | Drought gradient experiment

The site-based drought gradient experiment was conducted in an alpine meadow in Hongyuan county ($32^\circ 48' \text{N}$, $102^\circ 33' \text{E}$), located in the eastern of Qinghai-Tibetan Plateau. The topography of study site is relatively flat with an elevation of 3,500 m. Long-term (1961–2013) mean annual temperature was 1.5°C , with mean monthly temperature changing from -9.7°C in January to 11.1°C in July. Long-term mean annual precipitation was 747 mm, with 77.0% occurring during the growing season (May–September). The soil is classified as Mat Grygelic Cambisol by Chinese Soil Taxonomy Research Group (1995). The meadow community at our experimental site is dominated by *Deschampsia caespitosa* (Linn.) Beauv., *Kobresia setchwanensis* Hand.-Mazz., *Carex schneideri* Nelmes and *Polygonum viviparum* (Linn.) var. *angustum*. The vegetation cover of this grassland is over 90% (Song et al., 2017). During the experiment, the growing-season precipitation was 439.7, 445.2 and 500.4 mm in 2015, 2016 and 2017 respectively.

The precipitation gradient experiment was established in May 2015, using a randomized complete block design. There were five levels of precipitation (1/12 P, 1/4 P, 1/2 P, 3/4 P, and P, where P is the annual precipitation; Figure S6). Each plot was 2 × 1.5 m with five replicates and 2 m between the neighbouring plots. We used combinations of water catchments and rainout shelters to achieve the varying levels of precipitation. The rainout shelter was a fixed location shelter with a roof consisting of curved bands of transparent acrylic, which can intercept different amounts of rainfall while minimally changing other environment variables (Yahdjian & Sala, 2002). We installed fibreglass plates to a depth of 40 cm in the soil column surrounding the plots to hydrologically isolate the plots (Zavaleta et al., 2003). There was no change of precipitation frequency or timing in our design.

ANPP was estimated annually during peak above-ground plant biomass (August 6–10) using a 0.1 × 1 m quadrat randomly placed in each plot while avoiding overlapping quadrats during the 3 years. To reduce potential edge effects, we sampled at least 30 cm inside the border of each plot. All living plants were sorted to species, oven-dried at 65°C for 48 hr, and then weighed.

Plant species richness (number of species) was recorded in the same quadrat as the ANPP measurements each year. Species asynchrony was quantified as.

$$1 - \varphi_b = 1 - \sigma^2 / \left(\sum_{i=1}^S \sigma_{bi}^2 \right), \quad (4)$$

where φ_b is species synchrony, σ^2 is the variance of ecosystem ANPP and σ_{bi} is the standard deviation of ANPP of species i in a community with S species over the years 2015–2017 (Loreau & de Mazancourt, 2008).

We define resistance (Ω) as.

$$\Omega = Y_c / |Y_D - Y_c|, \quad (5)$$

where Y_D and Y_c are the ecosystem productivity during drought treatment and the control plot respectively. Resistance indicates maintaining productivity near control levels during drought treatment. For example, if productivity is reduced during a drought to half compared with the control, then $\Omega = 2$ (Isbell et al., 2015).

2.4 | Statistical analysis

In the meta-analysis, we constructed linear mixed-effects models with R package 'lme4' (Sarkar, 2008) to assess the contribution of RR to precipitation changes, with precipitation levels as fixed effects and the studies as random effects. One-way ANOVA was used to statistically compare the difference of RR between the extreme treatments and the non-extreme treatments.

In the precipitation gradient experiment, mixed linear models across all response variables were conducted with precipitation levels as fixed effects and block as random effects. A one-way ANOVA was also used to test the effects of precipitation on ANPP, species asynchrony across the 3 years. Linear regression analyses were used to evaluate the relationships of resistance and species asynchrony.

All statistics were performed through SPSS software package (SPSS 19.0 for windows, SPSS Inc., Chicago, IL).

Structural equation modelling (SEM) was performed using AMOS 17.0 (Amos Development, Spring House, PA) to estimate the effects of the precipitation treatments on the resistance of ecosystem productivity to drought. Following the concept of ecosystem resistance, and results from ANOVA and linear regression, we proposed an a priori model of hypothesized relationships within a path diagram (Figure S7), allowing a causal interpretation of the model outputs. Specifically, we constructed an a priori conceptual structural equation model depicting the effects of precipitation amount, species asynchrony, species richness, and dominant species change on ecosystem resistance. Data were fitted to the models applying the maximum likelihood estimation method. Adequate model fits are indicated by a non-significant χ^2 test ($p > 0.05$) and Akaike information criteria (AIC). We obtained the final SEM, which yielded a non-significant χ^2 test and the lowest AIC, by simplifying the initial SEM via eliminating non-significant pathways (Grace, 2006).

3 | RESULTS

The historical data analysis indicated that there was little co-occurrence of extremely low GPP and extreme drought (Figure 1). On average across all the grids in the world, 15.1% of extreme drought events occurred with extremely low GPP at the 5th percentile (Figure 1a) and 20.4% of extreme drought co-occurred with extremely low GPP at the 10th percentile (Figure 1b). The correlations were greater at the 10th percentile than that at the 5th percentile, because the former included more extreme events and possible correlations (Figure S8). Globally, the northern mid-latitude had stronger correlations between extremely low GPP and extreme drought than other regions. There were no significant differences of response coefficient between ecosystem types (Figure S9). The analysis also showed both indices of extreme drought (historical precipitation deficit and SPEI) had similar correlations with extreme low GPP (5th percentile; Figure S10).

The meta-analysis study revealed that there were no significant differences between the responses to extreme drought (>30% reductions in precipitation) versus non-extreme drought treatments (Figure 2a). Despite alternative groupings for precipitation reduction thresholds, we found no significant difference between groups (Figure S11). To assess the potential influence of confounding effects of treatment duration, we examined the effect of treatment duration on RRs but found no significant relationships across the drought experiments ($p = 0.99$, Figure 2b).

We also analysed the relationships between the RR of ANPP and other covariates across the study sites, including soil C and clay content, mean annual temperature and annual precipitation, yet only annual temperature was significantly correlated with the RRs (Figure S12). Next we examined the effect of temperature as a covariate in the model and used partial regression to analyse the

FIGURE 1 Global response coefficient between the historical extreme low gross primary productivity and extreme drought at the 5th (a) and 10th (b) percentile [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

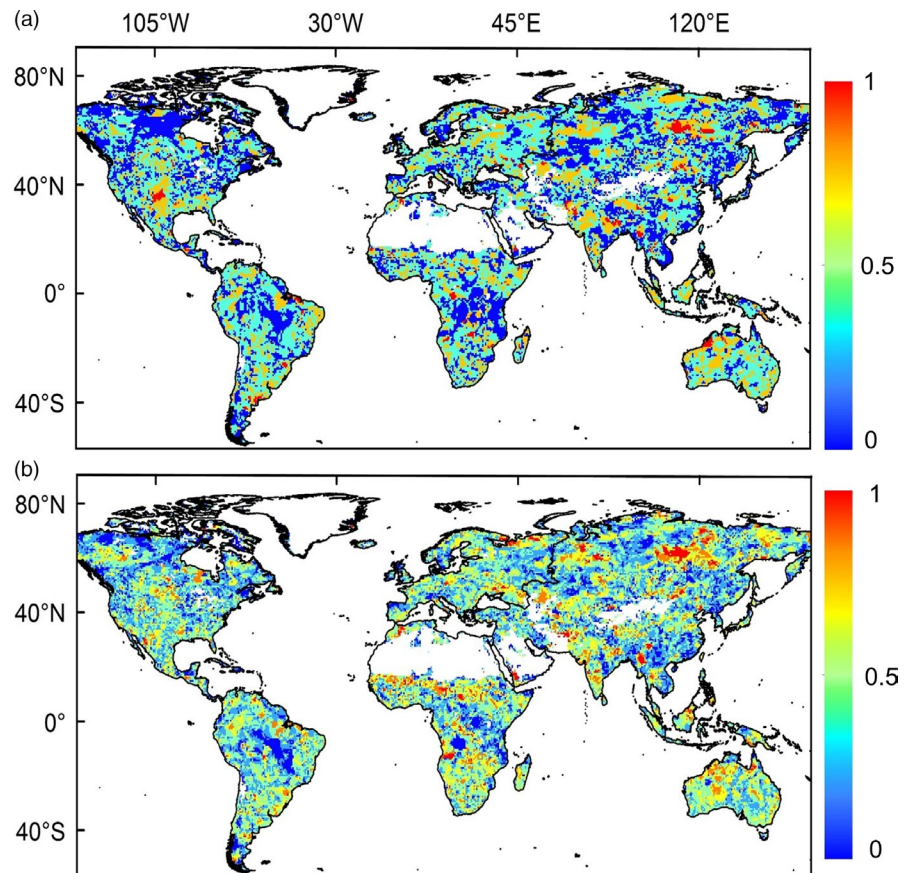
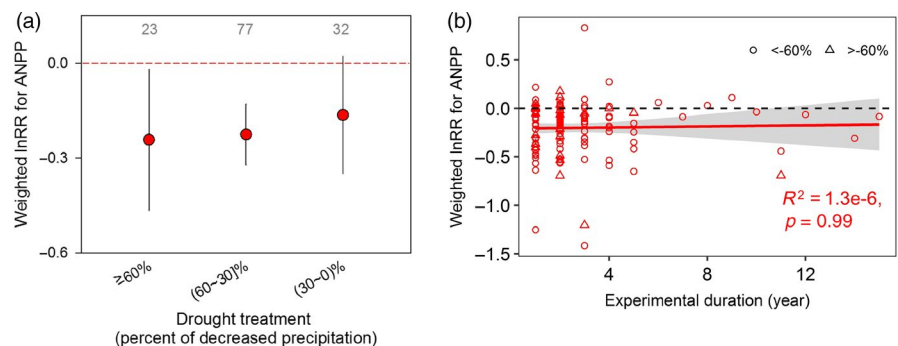


FIGURE 2 Global meta-analysis of the response ratio (RR) of above-ground net primary productivity (ANPP) under different precipitation reductions in manipulative precipitation experiments (a) and different experimental duration (b). The sample size for each group is shown in the top of panel a [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]



relationship between the RRs and drought magnitudes. These results also showed that drought magnitudes did not affect the RRs of ANPP (Figure S13).

To further explore what magnitude of drought could elicit an extreme ecological response, we conducted a drought gradient experiment. Four treatments, 1/12 P, 1/4 P, 1/2 P, and 3/4 P, were considered as extreme drought treatments based on the frequency distribution of long-term annual precipitation (below the 5th percentile; Figure 3a). Compared with the control, only the most extreme drought treatment (1/12 P) significantly decreased ANPP by 26.7% across the 3 years ($p < 0.05$, Figure 3b), while there were no significant differences between other extreme treatments and the control. When we examined the responses in each year, we found the 1/12 P treatment significantly decreased ANPP from 2016, 1 year after the treatment (Figure 4a). To identify the potential drivers of the treatment

effects on ANPP, we examined soil water content (SWC) across the precipitation treatments. The results showed that SWC fell below the wilting point for 0, 17 and 42 days under the 1/12 P treatment in 2015, 2016, and 2017 respectively, based on the study site soil type (Saxton & Rawls, 2006; Shui et al., 2017). SWC under the 1/4 P treatment fell below the wilting point for 14 days only in 2017. All other treatments did not result in SWC below the wilting point in any year (Figure S14).

To explore why only the most extreme drought treatment triggered the significant responses of ecosystem, we further examined the changes in species composition and community structure. We found that species richness was significantly decreased only in the 1/12 P treatment from 2016, coinciding with a negative response of ANPP. Specifically, species richness decreased by 19.8% in 2016 and 25.7% in 2017 (both $p < 0.01$, Figure 4b). The biomass of two dominant species, *D. caespitosa* (Linn.) Beauv. and *P. viviparum*

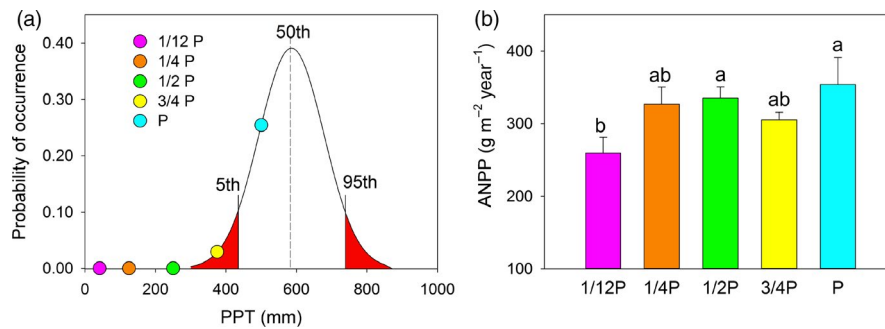


FIGURE 3 The extremity of precipitation treatment based on an estimated probability function calculated from 52 years of growing season precipitation at the study site (a). Variation in above-ground net primary productivity (ANPP) under different extremity drought treatments across the 3 years (b), values are $M \pm SE$. Different letters indicate statistically significant difference between treatments at $p < 0.05$ [Colour figure can be viewed at wileyonlinelibrary.com]

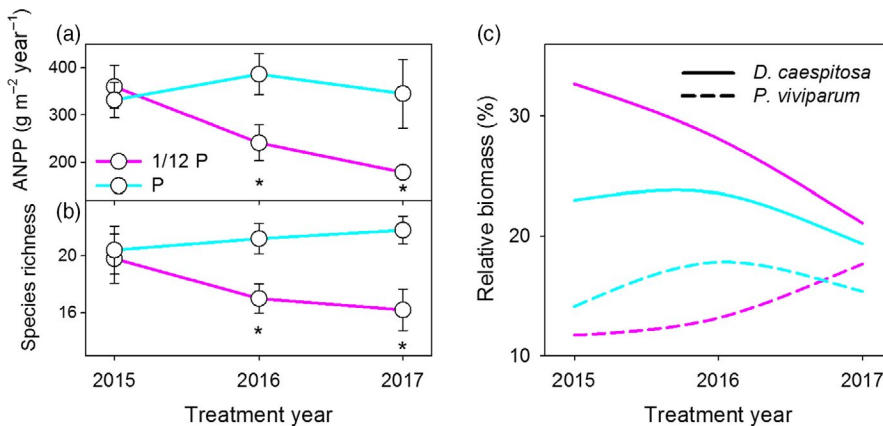


FIGURE 4 Above-ground net primary productivity (ANPP) (a), species richness (b) and relative biomass of two dominant species *Deschampsia caespitosa* (Linn.) Beauv. (solid line) and *Polygonum viviparum* (Linn.) var. *angustum* (dotted line) (c) under the most extreme precipitation treatment (1/12 P, pink) and the control (P, green) from 2015 to 2017. Asterisks indicate statistically significant difference at $p < 0.05$ [Colour figure can be viewed at wileyonlinelibrary.com]

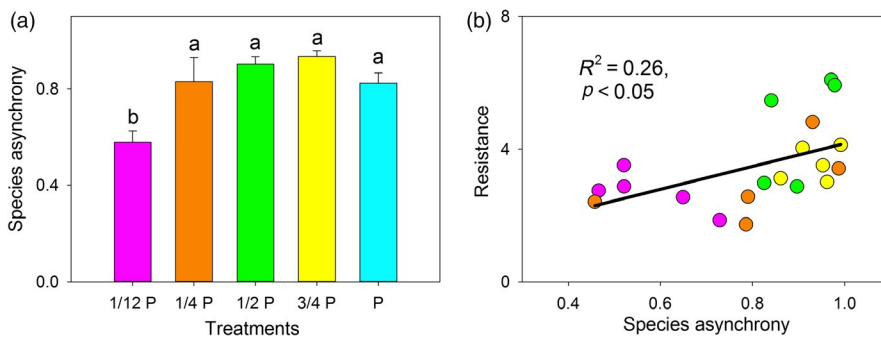


FIGURE 5 Variation in species asynchrony under treatments across the 3 years (a), values are $M \pm SE$. Different letters indicate statistically significant difference between treatments at $p < 0.05$. Relationships of species asynchrony and ecosystem resistance across the 3 years (b) [Colour figure can be viewed at wileyonlinelibrary.com]

(Linn.) var. *angustum*, which accounted for 37.7% of the total ANPP, responded oppositely to 1/12 P treatment from 2015 to 2017 (Figure 4c). Under 1/12 P treatment, the relative biomass of *P. viviparum* increased while that of *D. caespitosa* decreased dramatically. However, the relative biomass of these two species under the control plots remained relatively stable over the 3 years (Figure 4c), suggesting species reordering under 1/12 P treatment.

After 3 years of treatments, species asynchrony significantly decreased by 29.9% under the 1/12 P treatment, while other treatments had no differences (Figure 5a). Moreover, species asynchrony increased positively with resistance of ecosystem productivity ($p < 0.05$; Figure 5b). The SEM indicated that extreme drought significantly reduced species richness (standardized effect size: 0.68; $p < 0.001$) and species asynchrony (standardized effect size: 0.72; $p < 0.001$), leading to decreased resistance of ecosystem productivity to extreme drought (Figure 6).

4 | DISCUSSION

For this study, we combined historical data mining, global meta-analysis, and a manipulative experiment to comprehensively examine when extreme drought elicits extreme ecological responses and why. This complementary approach revealed large-scale patterns and improved our mechanistic understanding. As we hypothesized, observed historical extreme drought did not consistently elicit extreme GPP responses. Furthermore, experimentally imposed extreme droughts did not result in extreme ecological responses. The drought gradient experiment revealed that only the most extreme drought treatment (1/12 P), rather than other extremities, significantly decreased ANPP. Under the 1/12 P treatment, SWC was below a critical wilting point threshold, leading to decreased species asynchrony, lower species richness, and a reordering of dominant species

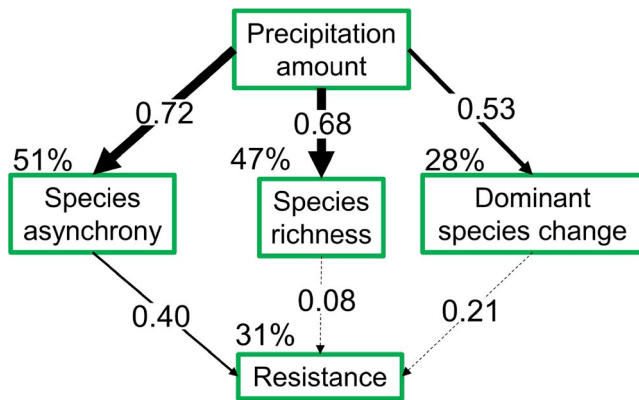


FIGURE 6 The structural equation modelling (SEM) for the effects of precipitation change on species asynchrony, species richness, dominant species change (standard deviation of dominant species biomass across 3 years) and resistance. The final SEM fitted the data well: $\chi^2 = 1.044$, $df = 4$, $p = 0.903$. Width of the arrows is proportional to the strength of the relationship, and numbers at arrows are standardized path coefficients. Dashed and solid arrows indicate $p > 0.1$ and $p < 0.1$, respectively. Black arrows indicate positive relationship. Percentages on top of response variables indicate the proportion of variance explained by relationships with other variables (R^2) [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/1365-2745.13226)]

composition. Based on these findings, a conceptual framework proposed here explains ecosystem response to precipitation changes at different stages (Figure 7). This study reveals biological mechanisms driving ecosystem responses to extreme precipitation and suggests the critical importance of species asynchrony and diversity in determining ecosystem resistance to extreme climate events.

4.1 | The correlation between extreme climate and extreme ecosystem responses

Historical observations revealed that the majority of extreme drought events at both the 5th and 10th percentile did not co-occur with extremely low GPP. The results verified our initial hypothesis that extreme drought does not necessarily cause extreme GPP responses. There are three potential reasons that may account for this poor correlation between extreme GPP and precipitation.

First, the magnitude of the naturally occurring droughts may have not been sufficient to trigger an ecological response. Particularly in some ecosystems, which are controlled by long-lived species, species replacement seldom changes under extreme climate perturbations (Lloret, Escudero, Iriondo, Martínez-Vilalta, & Valladares, 2012). Additionally, we found that all vegetation types showed low correlation between historically extreme precipitation and GPP, and there was no significant difference between them. In other ecosystems, plant communities may be altered following extreme climate events, though ecosystem function may recover completely due to rapid demographic responses by dominant species, which compensate for the loss of others (Dreesen, Boeck, Janssens, & Nijs, 2012; Hoover et al., 2014b; Jentsch et al., 2011). Thus, understanding plant community dynamics is critical for revealing ecosystem responses to extreme precipitation.

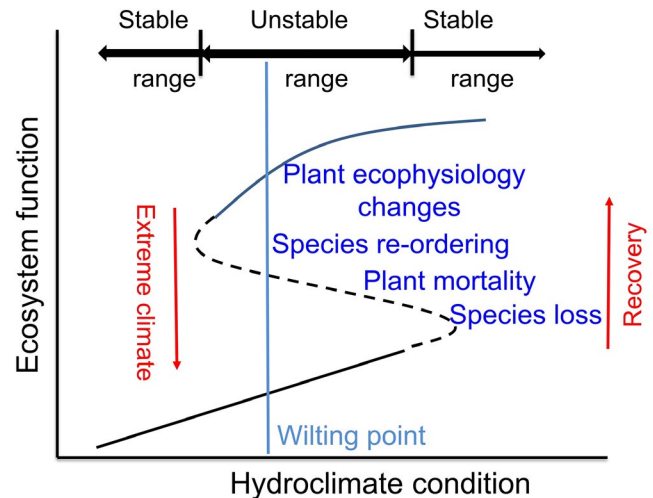


FIGURE 7 A mechanistic framework for assessing ecosystem function responses to hydroclimate condition. Precipitation extremes primarily cause individual-level plant ecophysiology responses first and then result in species reordering, loss of asynchrony and eventually the species mortality and loss, with a consequent extreme response of ecosystem function and an unstable state [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/1365-2745.13226)]

Second, the characteristics of precipitation, (e.g. event size, intensity, and frequency), also affect soil water availability, and thus the magnitude of a drought event can be driven by both precipitation pattern and amount (Brunetti, Maugeri, Nanni, & Navarra, 2002; Liu et al., 2013). For example, it was reported that an index of extreme water availability, calculated by precipitation and potential evapotranspiration, rather than precipitation amount alone, dominates global interannual variability in GPP (Zscheischler et al., 2014). These confounding factors may account for the poor correlation between precipitation and extreme ecological responses. While we focused on the relationship between extreme drought (low precipitation) and extreme GPP at the annual timescale in this study, the potential impacts of additional climate factors and precipitation indices (drought timing, intensity, frequency) warrant further investigation.

Third, the traditional percentile-based approach, which defines extreme as the lower 5th or 10th percentile of distribution, may not accurately quantify extreme drought or ecological responses (Hoover et al., 2014b). For example, the relative deviation in precipitation levels from the average, which is necessary to meet a statistical definition of extremity, was lower and relatively constant in high precipitation areas, which may not cause extreme ecological responses (Knapp et al., 2015). However, in arid ecosystems, the percent deviations from an average year were greater than more mesic ecosystems, and perhaps more likely to cause an extreme response. Therefore, for a given year to be statistically classified as one having extreme precipitation, the magnitude of extremity differs significantly across different climates and ecosystems. In addition, the choice of the percentile threshold may impact the conclusions. Our analysis showed that the percentage of correlations increased under higher percentiles, because larger percentile will include more extreme events and possible correlation (Figure S8). This suggests that we need to develop more

sophisticated and dynamic statistical techniques to define extreme events for different ecosystem aspects (Niu et al., 2014).

4.2 | What magnitude of drought elicits an extreme ecological response and why?

While historical data mining is well-suited for explaining ecological responses to a changing climate, it is unable to provide a mechanistic understanding of these responses due to several potential confounding factors. Thus, we used a manipulative experiment to address this question. Although there were four extreme drought treatments in our study, only the most extreme drought treatment (1/12 P) significantly reduced ANPP (Figure 3). This begged the question: Why did only the most extreme drought treatment cause an extreme response? Below, we propose the potential biological mechanisms that may be driving this response.

Extreme drought may be more likely to decrease productivity if there are changes in species diversity. Biodiversity, in particular, plant species richness, plays a critical role in ecosystem productivity and stability (Hautier et al., 2015; Lehman & Tilman, 2000; Loreau & de Mazancourt, 2013). Species loss due to extreme drought, particularly for annual forbs and exotic annual grasses, led to a reduction in productivity in some experiments (Copeland et al., 2016; Tilman & El Haddi, 1992). Further, global meta-analyses overwhelmingly confirmed that decreased plant biodiversity can reduce plant productivity (Balvanera et al., 2006; Cardinale et al., 2011). Additionally, local species loss is a major driver of changes in ecosystem production and leads to remarkable ecological consequences under global change stressors (e.g., elevated CO₂ and nitrogen addition) (Hooper et al., 2012). Species richness decreased under 1/12 P treatment over the 3 years of our study (Figure 4). Compared with the control, approximately four and six species were lost in 2016 and 2017, respectively. However, other treatments did not reduce species richness, with non-significant changes in ANPP. Without species loss, the decreased growth of some species may be compensated by the increased growth of other drought-adapted species. For example, the decreased biomass of two species, *D. caespitosa* and *C. schneideri* under the 3/4 P treatment was compensated by biomass increases of three species: *P. viviparum*, *Gentiana sino-ornata* and *Carum carvi*.

Community reordering is also a major mechanism that can decrease productivity under extreme climate events (Cavin, Mountford, Peterken, & Jump, 2013). In this study, divergence in the relative biomass of the two dominant species under the 1/12 P treatment (Figure 5) leads to a reordering of the dominant species. Decreases in soil water availability and nutrient availability under the most extreme drought treatment may have increased competition between species, which disrupted community balance (Bardgett, Manning, Morriën, & Vries, 2013; Hoover, Knapp, & Smith, 2014a). A multiyear manipulative drought study also verified that dominant species played an important role in the maintenance of ecosystem stability (Evans, Byrne, Lauenroth, & Burke, 2011). The reduction in dominant species typically corresponds to an increase in subordinate species, which prefer to colonize arid soils. This suggested that

species reordering could weaken ecosystem stability under extreme climate events.

A large decrease in species asynchrony under extreme drought contributed substantially to the reduction in ANPP. Significant reductions in species asynchrony can leave ecosystems more vulnerable to extreme climate events (Figure 7) (Ma et al., 2017; Wilcox et al., 2017; Zhang, Loreau, He, Zhang, & Han, 2017). Asynchrony buffers community productivity to environmental fluctuations as reduced biomass of vulnerable species is compensated by growth of other species (Hautier et al., 2014; Loreau & de Mazancourt, 2008; Yachi & Loreau, 1999). When species fluctuations are not synchronized, asynchrony increases niche partitioning or facilitation, which leads to a high resistance of productivity to environment changes (Xu et al., 2014). Our results suggest that species asynchrony may be an important mechanism for ecosystem resistance to extreme drought.

4.3 | Implications and recommendations for future studies

Understanding the effects of climate extremes on carbon fluxes is critical for accurate prediction of changes in terrestrial carbon sinks (Reichstein et al., 2013). The weak correlations between historical extreme GPP and extreme drought globally do not necessarily translate to null responses at the species-level. Community dynamics (e.g. plant ecophysiological changes, species reordering, plant mortality and species loss), play a key role in ecosystem responses to extreme precipitation (Figure 7). Reduced ANPP under the extreme drought treatment was most likely due to reductions in plant growth when SWC was below the wilting point (Kramer & Boyer, 1995). This response may vary from plant to plant, leading to changes in community structure. Species that are more sensitive to extreme drought may cross physiological thresholds, eventually leading to mortality and species loss. Modellers should incorporate such community dynamics into their models, to better simulate ecosystem productivity during climate extremes (Reichstein et al., 2013).

Additional research is needed to further understand the relationship between extreme events and ecosystem functions. First, drought gradient experiments should be conducted in a wider range of ecosystem types. These experiments can help to reveal how extreme climate sensitivity varies across a broad range of terrestrial ecosystems. Rather than traditional site-based experiments, coordinated distributed drought networks can also improve our understanding the differential sensitivities to extreme climate at both regional and global scales (Beier et al., 2012; Knapp et al., 2017; Smith, Wilcox, Power, Tissue, & Knapp, 2017). Second, since natural extreme climate events typically occur in combination naturally, it is also essential to assess interactions between different types of extreme climate (Smith, 2011). New experiments should simulate the interactions between extreme precipitation and other key factors involved with global climate change (such as elevated CO₂, warming). Aside from the climate change factors, other drivers related to experimental responses, such as changes in ecological traits, species and genotype diversity and disturbance regimes should also be

considered, as these drivers may interact with climate extremities to induce the most extreme ecological responses (Smith et al., 2009). Finally, studies that combine data mining or experimental results with simulation models might provide bring new insights and more accurate predictions of extreme climate effects (Beier et al., 2012).

5 | CONCLUSIONS

For this study, we demonstrated that extreme drought events do not necessarily elicit extremely low GPP, with a small fraction of historical extremely low GPP correlated with extreme drought globally. Our field experiment indicated that only the 1/12 P extreme drought treatment decreased ANPP, due to a decline of species asynchrony and community diversity, as well as reordering between the two dominant species. These findings suggested the critical importance of species asynchrony and diversity in the determination of ecosystem resistance to extreme climate events. Future efforts with additional mechanistic focus associated with extremity gradient experiments are required to better understand different ecosystem responses to extreme climate events.

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AUTHORS' CONTRIBUTIONS

S.N. designed the study. F.Z., Q.Q., F.M. and Q.Z. conducted the field experiment and analysed the data. F.Z. wrote the first draft of the paper. S.N., F.Z., D.L.H., D.T. and Q.Z. commented on the details of the manuscript drafts. All authors provided input to the drafting and final version of the manuscript.

DATA AVAILABILITY STATEMENT

Global GPP data can be retrieved from Data Portal of the Max Planck Institute for Biogeochemistry (<https://www.bgc-jena.mpg.de/geodb/projects/Data.php>). Global annual precipitation data can be retrieved from CRU-High-resolution gridded datasets (<https://crudata.uea.ac.uk/cru/data/hrg/>). The experimental data can be obtained from the Dryad Digital Repository <https://doi.org/10.5061/dryad.rj6h2q7> (Zhang et al., 2019).

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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